CANONICAL TATE DETERMINANT-TO-DUALIZING ISOMORPHISM FOR QUASI-FINITE SYNTOMIC MAPS

THE STACKS PROJECT AUTHORS

Editorial note. In the fibre-product display the second tensor product is over A_2, as fixed by both its factors and the following base-change identity. The printed A_1 subscript is retained and disclosed.

Editorial note. In the Noetherian approximation paragraph, the finite polynomial subalgebra notation is missing a closing bracket; no mathematical variable or hypothesis is changed.

Editorial note. A Garel differential-form construction in the adjacent chapter uses a related descent pattern but a different map with additional exterior-form obligations; it is not a second proof of this determinant/dualizing isomorphism.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Determinant of the naive cotangent complex, dualizing modules for quasi-finite maps, trace base change, flat finite-presentation patching over fibre products, and finite-presentation limit descent. The local different formula and universal syntomic family are retained author support.

Source reference. Original lemma 069K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8044–8044. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01UF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2408–2408. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 00SP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1047–1047. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00SW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1054–1054. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00PK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 934–934. Stable Stacks tag for an expressly allowed

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foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00SY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1056–1056. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01UE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2407–2407. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FK3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19448–19448. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00SV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1053–1053. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BVE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14907–14907. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 080Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10190–10190. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BVF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14908–14908. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0BVG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14909–14909. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00OY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 912–912. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AX0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13738–13738. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0A71: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12864–12864. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DW9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17389–17389. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DWA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17390–17390. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00QA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 960–960. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3241–3241. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0624: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7783–7783. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07D2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9397–9397. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 068M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8011–8011. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 062H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7796–7796. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BW6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14934–14934. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BVZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14927–14927. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BVT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14921–14921. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BW9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14937–14937. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0H9B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21521–21521. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07RU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9879–9879. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08KQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10893–10893. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01ZM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2599–2599. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0C3L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15196–15196. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 094M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11555–11555. Stable Stacks tag for an expressly

allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FJX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19442–19442. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FK0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19445–19445. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0FJY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19443–19443. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0BVJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14912–14912. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original equation 0BVB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14904–14904. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BT9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14832–14832. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Proposition 0.1. *There exists a unique rule that to every locally quasi-finite syntomic morphism of locally Noetherian schemes $Y \to X$ assigns an isomorphism*

$$c_{Y/X} : \det(NL_{Y/X}) \longrightarrow \omega_{Y/X}$$

satisfying the following two properties

(1) *the section $\delta(NL_{Y/X})$ is mapped to $\tau_{Y/X}$, and*
(2) *the rule is compatible with restriction to opens and with base change.*

1. The Tate map

In this section we produce an isomorphism between the determinant of the relative cotangent complex and the relative dualizing module for a locally quasi-finite syntomic morphism of locally Noetherian schemes. Following [1, 1.4.4] we dub the isomorphism the Tate map. Our approach is to avoid doing local calculations as much as is possible.

Let $Y \to X$ be a locally quasi-finite syntomic morphism of schemes. We will use all the equivalent conditions for this notion given in Lemma 1.2 without further

mention in this section. In particular, we see that $NL_{Y/X}$ is a perfect object of $D(\mathcal{O}_Y)$ with tor-amplitude in $[-1, 0]$. Thus we have a canonical invertible module $\det(NL_{Y/X})$ on Y and a global section

$$\delta(NL_{Y/X}) \in \Gamma(Y, \det(NL_{Y/X}))$$

See Derived Categories of Schemes, Lemma 0FJX. Suppose given a commutative diagram of schemes

$$\begin{array}{ccc} Y' & \xrightarrow[b]{} & Y \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

whose vertical arrows are locally quasi-finite syntomic and which induces an isomorphism of Y' with an open of $X' \times_X Y$. Then the canonical map

$$Lb^* NL_{Y/X} \longrightarrow NL_{Y'/X'}$$

is a quasi-isomorphism by More on Morphisms, Lemma 0FK0. Thus we get a canonical isomorphism $b^* \det(NL_{Y/X}) \to \det(NL_{Y'/X'})$ which sends the canonical section $\delta(NL_{Y/X})$ to $\delta(NL_{Y'/X'})$, see Derived Categories of Schemes, Remark 0FJY.

Remark 1.1. Let $Y \to X$ be a locally quasi-finite syntomic morphism of schemes. What does the pair $(\det(NL_{Y/X}), \delta(NL_{Y/X}))$ look like locally? Choose affine opens $V = \mathrm{Spec}(B) \subset Y$, $U = \mathrm{Spec}(A) \subset X$ with $f(V) \subset U$ and an integer n and $f_1, \ldots, f_n \in A[x_1, \ldots, x_n]$ such that $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$. Then

$$NL_{B/A} = \Big((f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2 \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} B\mathrm{d}x_i \Big)$$

and $(f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2$ is free with generators the classes $\overline{f}_i$. See proof of Lemma 1.2. Thus $\det(L_{B/A})$ is free on the generator

$$\mathrm{d}x_1 \wedge \ldots \wedge \mathrm{d}x_n \otimes (\overline{f}_1 \wedge \ldots \wedge \overline{f}_n)^{\otimes -1}$$

and the section $\delta(NL_{B/A})$ is the element

$$\delta(NL_{B/A}) = \det(\partial f_j/\partial x_i) \cdot \mathrm{d}x_1 \wedge \ldots \wedge \mathrm{d}x_n \otimes (\overline{f}_1 \wedge \ldots \wedge \overline{f}_n)^{\otimes -1}$$

by definition.

Let $Y \to X$ be a locally quasi-finite syntomic morphism of locally Noetherian schemes. By Remarks 0BVG and 0BVJ we have a coherent $\mathcal{O}_Y$-module $\omega_{Y/X}$ and a canonical global section

$$\tau_{Y/X} \in \Gamma(Y, \omega_{Y/X})$$

which affine locally recovers the pair $\omega_{B/A}, \tau_{B/A}$. By Lemma 1.5 the module $\omega_{Y/X}$ is invertible. Suppose given a commutative diagram of locally Noetherian schemes

$$\begin{array}{ccc} Y' & \xrightarrow[b]{} & Y \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

whose vertical arrows are locally quasi-finite syntomic and which induces an isomorphism of Y' with an open of $X' \times_X Y$. Then there is a canonical base change map

$$b^* \omega_{Y/X} \longrightarrow \omega_{Y'/X'}$$

which is an isomorphism mapping $\tau_{Y/X}$ to $\tau_{Y'/X'}$. Namely, the base change map in the affine setting is (0BVB), it is an isomorphism by Lemma 0BVF, and it maps $\tau_{Y/X}$ to $\tau_{Y'/X'}$ by Lemma 0BT9 part (1).

Lemma 1.2. *Let $f : Y \to X$ be a morphism of schemes. The following are equivalent*

(1) *f is locally quasi-finite and syntomic,*
(2) *f is locally quasi-finite, flat, and a local complete intersection morphism,*
(3) *f is locally quasi-finite, flat, locally of finite presentation, and the fibres of f are local complete intersections,*
(4) *f is locally quasi-finite and for every $y \in Y$ there are affine opens $y \in V = \mathrm{Spec}(B) \subset Y$, $U = \mathrm{Spec}(A) \subset X$ with $f(V) \subset U$ an integer n and $h, f_1, \ldots, f_n \in A[x_1, \ldots, x_n]$ such that $B = A[x_1, \ldots, x_n, 1/h]/(f_1, \ldots, f_n)$,*
(5) *for every $y \in Y$ there are affine opens $y \in V = \mathrm{Spec}(B) \subset Y$, $U = \mathrm{Spec}(A) \subset X$ with $f(V) \subset U$ such that $A \to B$ is a relative global complete intersection of the form $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$,*
(6) *f is locally quasi-finite, flat, locally of finite presentation, and $NL_{Y/X}$ has tor-amplitude in $[-1, 0]$, and*
(7) *f is flat, locally of finite presentation, $NL_{Y/X}$ is perfect of rank 0 with tor-amplitude in $[-1, 0]$,*

Proof. The equivalence of (1) and (2) is More on Morphisms, Lemma 069K. The equivalence of (1) and (3) is Morphisms, Lemma 01UF.

If $A \to B$ is as in (4), then $B = A[x, x_1, \ldots, x_n]/(xh - 1, f_1, \ldots, f_n)$ is a relative global complete intersection by see Algebra, Definition 00SP. Thus (4) implies (5). It is clear that (5) implies (4).

Condition (5) implies (1): by Algebra, Lemma 00SW a relative global complete intersection is syntomic and the definition of a relative global complete intersection guarantees that a relative global complete intersection on n variables with n equations is quasi-finite, see Algebra, Definition 00SP and Lemma 00PK.

Either Algebra, Lemma 00SY or Morphisms, Lemma 01UE shows that (1) implies (5).

More on Morphisms, Lemma 0FK3 shows that (6) is equivalent to (1). If the equivalent conditions (1) – (6) hold, then we see that affine locally $Y \to X$ is given by a relative global complete intersection $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$ with the same number of variables as the number of equations. Using this presentation we see that

$$NL_{B/A} = \Big((f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2 \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} B\mathrm{d}x_i\Big)$$

By Algebra, Lemma 00SV the module $(f_1, \ldots, f_n)/(f_1, \ldots, f_n)^2$ is free with generators the congruence classes of the elements $f_1, \ldots, f_n$. Thus $NL_{B/A}$ has rank 0 and so does $NL_{Y/X}$. In this way we see that (1) – (6) imply (7).

Finally, assume (7). By More on Morphisms, Lemma 0FK3 we see that f is syntomic. Thus on suitable affine opens f is given by a relative global complete intersection $A \to B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$, see Morphisms, Lemma 01UE. Exactly as above we see that $NL_{B/A}$ is a perfect complex of rank $n - m$. Thus $n = m$ and we see that (5) holds. This finishes the proof. □

Lemma 1.3. *Invertibility of the relative dualizing module.*

(1) *If $A \to B$ is a quasi-finite flat homomorphism of Noetherian rings, then $\omega_{B/A}$ is an invertible B-module if and only if $\omega_{B \otimes_A \kappa(\mathfrak{p})/\kappa(\mathfrak{p})}$ is an invertible $B \otimes_A \kappa(\mathfrak{p})$-module for all primes $\mathfrak{p} \subset A$.*
(2) *If $Y \to X$ is a quasi-finite flat morphism of Noetherian schemes, then $\omega_{Y/X}$ is invertible if and only if $\omega_{Y_x/x}$ is invertible for all $x \in X$.*

Proof. Proof of (1). As $A \to B$ is flat, the module $\omega_{B/A}$ is A-flat, see Lemma 0BVE. Thus $\omega_{B/A}$ is an invertible B-module if and only if $\omega_{B/A} \otimes_A \kappa(\mathfrak{p})$ is an invertible $B \otimes_A \kappa(\mathfrak{p})$-module for every prime $\mathfrak{p} \subset A$, see More on Morphisms, Lemma 080Q. Still using that $A \to B$ is flat, we have that formation of $\omega_{B/A}$ commutes with base change, see Lemma 0BVF. Thus we see that invertibility of the relative dualizing module, in the presence of flatness, is equivalent to invertibility of the relative dualizing module for the maps $\kappa(\mathfrak{p}) \to B \otimes_A \kappa(\mathfrak{p})$.

Part (2) follows from (1) and the fact that affine locally the dualizing modules are given by their algebraic counterparts, see Remark 0BVG. □

Lemma 1.4. *Let k be a field. Let $B = k[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$ be a global complete intersection over k of dimension 0. Then $\omega_{B/k}$ is invertible.*

Proof. By Noether normalization, see Algebra, Lemma 00OY we see that there exists a finite injection $k \to B$, i.e., $\dim_k(B) < \infty$. Hence $\omega_{B/k} = \mathrm{Hom}_k(B, k)$ as a B-module. By Dualizing Complexes, Lemma 0AX0 we see that $R\,\mathrm{Hom}(B, k)$ is a dualizing complex for B and by Dualizing Complexes, Lemma 0A71 we see that $R\,\mathrm{Hom}(B, k)$ is equal to $\omega_{B/k}$ placed in degree 0. Thus it suffices to show that B is Gorenstein (Dualizing Complexes, Lemma 0DW9). This is true by Dualizing Complexes, Lemma 0DWA. □

Lemma 1.5. *Let $f : Y \to X$ be a morphism of locally Noetherian schemes. If f satisfies the equivalent conditions of Lemma 1.2 then $\omega_{Y/X}$ is an invertible $\mathcal{O}_Y$-module.*

Proof. We may assume $A \to B$ is a relative global complete intersection of the form $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$ and we have to show $\omega_{B/A}$ is invertible. This follows in combining Lemmas 1.3 and 1.4. □

Example 1.6. Let $n \geq 1$ and $d \geq 1$ be integers. Let T be the set of multi-indices $E = (e_1, \ldots, e_n)$ with $e_i \geq 0$ and $\sum e_i \leq d$. Consider the ring

$$A = \mathbf{Z}[a_{i,E}; 1 \leq i \leq n, E \in T]$$

In $A[x_1, \ldots, x_n]$ consider the elements $f_i = \sum_{E \in T} a_{i,E} x^E$ where $x^E = x_1^{e_1} \ldots x_n^{e_n}$ as is customary. Consider the A-algebra

$$B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$$

Denote $X_{n,d} = \mathrm{Spec}(A)$ and let $Y_{n,d} \subset \mathrm{Spec}(B)$ be the maximal open subscheme such that the restriction of the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A) = X_{n,d}$ is quasi-finite, see Algebra, Lemma 00QA.

Lemma 1.7. *With notation as in Example 1.6 the schemes $X_{n,d}$ and $Y_{n,d}$ are regular and irreducible, the morphism $Y_{n,d} \to X_{n,d}$ is locally quasi-finite and syntomic, and there is a dense open subscheme $V \subset Y_{n,d}$ such that $Y_{n,d} \to X_{n,d}$ restricts to an étale morphism $V \to X_{n,d}$.*

Proof. The scheme $X_{n,d}$ is the spectrum of the polynomial ring A. Hence $X_{n,d}$ is regular and irreducible. Since we can write

$$f_i = a_{i,(0,\ldots,0)} + \sum\nolimits_{E \in T, E \neq (0,\ldots,0)} a_{i,E} x^E$$

we see that the ring B is isomorphic to the polynomial ring on $x_1, \ldots, x_n$ and the elements $a_{i,E}$ with $E \neq (0, \ldots, 0)$. Hence $\mathrm{Spec}(B)$ is an irreducible and regular scheme and so is the open $Y_{n,d}$. The morphism $Y_{n,d} \to X_{n,d}$ is locally quasi-finite and syntomic by Lemma 1.2. To find V it suffices to find a single point where $Y_{n,d} \to X_{n,d}$ is étale (the locus of points where a morphism is étale is open by definition). Thus it suffices to find a point of $X_{n,d}$ where the fibre of $Y_{n,d} \to X_{n,d}$ is nonempty

and étale, see Morphisms, Lemma 02GU. We choose the point corresponding to the ring map $\chi : A \to \mathbf{Q}$ sending f_i to $x_i^d - 1$. Then

$$B \otimes_{A,\chi} \mathbf{Q} = \mathbf{Q}[x_1, \ldots, x_n]/(x_1^d - 1, \ldots, x_n^d - 1)$$

which is a nonzero étale algebra over $\mathbf{Q}$. □

Lemma 1.8. *Let $f : Y \to X$ be a morphism of schemes. If f satisfies the equivalent conditions of Lemma 1.2 then for every $y \in Y$ there exist n, d and a commutative diagram*

$$\begin{array}{ccccc} Y & \longleftarrow & V & \longrightarrow & Y_{n,d} \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & U & \longrightarrow & X_{n,d} \end{array}$$

where $U \subset X$ and $V \subset Y$ are open with $y \in V$, where $Y_{n,d} \to X_{n,d}$ is as in Example 1.6, and where the square on the right hand side is cartesian.

Proof. By Lemma 1.2 we can choose U and V affine so that $U = \mathrm{Spec}(R)$ and $V = \mathrm{Spec}(S)$ with $S = R[y_1, \ldots, y_n]/(g_1, \ldots, g_n)$. With notation as in Example 1.6 if we pick d large enough, then we can write each g_i as $g_i = \sum_{E \in T} g_{i,E} y^E$ with $g_{i,E} \in R$. Then the map $A \to R$ sending $a_{i,E}$ to $g_{i,E}$ and the map $B \to S$ sending $x_i \to y_i$ give a cocartesian diagram of rings

$$\begin{array}{ccc} S & \longleftarrow & B \\ \uparrow & & \uparrow \\ R & \longleftarrow & A \end{array}$$

which proves the lemma. □

2. A formula for the different

In this section we discuss the material in [2, Appendix A] due to Tate. In our language, this will show that the different is equal to the Kähler different in the case of a flat, quasi-finite, local complete intersection morphism. First we compute the Noether different in a special case.

Lemma 2.1. [2, Appendix] *Let $A \to P$ be a ring map. Let $f_1, \ldots, f_n \in P$ be a Koszul regular sequence. Assume $B = P/(f_1, \ldots, f_n)$ is flat over A. Let $g_1, \ldots, g_n \in P \otimes_A B$ be a Koszul regular sequence generating the kernel of the multiplication map $P \otimes_A B \to B$. Write $f_i \otimes 1 = \sum g_{ij} g_j$. Then the annihilator of $\mathrm{Ker}(B \otimes_A B \to B)$ is a principal ideal generated by the image of $\det(g_{ij})$.*

Proof. The Koszul complex $K_\bullet = K(P, f_1, \ldots, f_n)$ is a resolution of B by finite free P-modules. The Koszul complex $M_\bullet = K(P \otimes_A B, g_1, \ldots, g_n)$ is a resolution of B by finite free $P \otimes_A B$-modules. There is a map of complexes

$$K_\bullet \longrightarrow M_\bullet$$

which in degree 1 is given by the matrix (g_{ij}) and in degree n by $\det(g_{ij})$. See More on Algebra, Lemma 0624. As B is a flat A-module, we can view $M_\bullet$ as a complex of flat P-modules (via $P \to P \otimes_A B$, $p \mapsto p \otimes 1$). Thus we may use both complexes to compute $\mathrm{Tor}_*^P(B, B)$ and it follows that the displayed map defines a quasi-isomorphism after tensoring with B. It is clear that $H_n(K_\bullet \otimes_P B) = B$. On the other hand, $H_n(M_\bullet \otimes_P B)$ is the kernel of

$$B \otimes_A B \xrightarrow{g_1, \ldots, g_n} (B \otimes_A B)^{\oplus n}$$

Since $g_1, \ldots, g_n$ generate the kernel of $B \otimes_A B \to B$ this proves the lemma. □

Lemma 2.2. *Let A be a ring. Let $n \geq 1$ and $h, f_1, \ldots, f_n \in A[x_1, \ldots, x_n]$. Set $B = A[x_1, \ldots, x_n, 1/h]/(f_1, \ldots, f_n)$. Assume that B is quasi-finite over A. Then*

1. *B is flat over A and $A \to B$ is a relative local complete intersection,*
2. *the annihilator J of $I = \mathrm{Ker}(B \otimes_A B \to B)$ is free of rank 1 over B,*
3. *the Noether different of B over A is generated by $\det(\partial f_i/\partial x_j)$ in B.*

Proof. Note that $B = A[x, x_1, \ldots, x_n]/(xh - 1, f_1, \ldots, f_n)$ is a relative global complete intersection over A, see Algebra, Definition 00SP. By Algebra, Lemma 00SW we see that B is flat over A.

Write $P' = A[x, x_1, \ldots, x_n]$ and $P = P'/(xh - 1) = A[x_1, \ldots, x_n, 1/h]$. Then we have $P' \to P \to B$. By More on Algebra, Lemma 07D2 we see that $xh-1, f_1, \ldots, f_n$ is a Koszul regular sequence in P'. Since $xh - 1$ is a Koszul regular sequence of length one in P' (by the same lemma for example) we conclude that $f_1, \ldots, f_n$ is a Koszul regular sequence in P by More on Algebra, Lemma 068M.

Let $g_i \in P \otimes_A B$ be the image of $x_i \otimes 1 - 1 \otimes x_i$. Let us use the short hand $y_i = x_i \otimes 1$ and $z_i = 1 \otimes x_i$ in $A[x_1, \ldots, x_n] \otimes_A A[x_1, \ldots, x_n]$ so that g_i is the image of $y_i - z_i$. For a polynomial $f \in A[x_1, \ldots, x_n]$ we write $f(y) = f \otimes 1$ and $f(z) = 1 \otimes f$ in the above tensor product. Then we have

$$P \otimes_A B/(g_1, \ldots, g_n) = \frac{A[y_1, \ldots, y_n, z_1, \ldots, z_n, \frac{1}{h(y)h(z)}]}{(f_1(z), \ldots, f_n(z), y_1 - z_1, \ldots, y_n - z_n)}$$

which is clearly isomorphic to B. Hence by the same arguments as above we find that $f_1(z), \ldots, f_n(z), y_1 - z_1, \ldots, y_n - z_n$ is a Koszul regular sequence in $A[y_1, \ldots, y_n, z_1, \ldots, z_n, \frac{1}{h(y)h(z)}]$. The sequence $f_1(z), \ldots, f_n(z)$ is a Koszul regular in $A[y_1, \ldots, y_n, z_1, \ldots, z_n, \frac{1}{h(y)h(z)}]$ by flatness of the map

$$P \longrightarrow A[y_1, \ldots, y_n, z_1, \ldots, z_n, \tfrac{1}{h(y)h(z)}], \quad x_i \longmapsto z_i$$

and More on Algebra, Lemma 062H. By More on Algebra, Lemma 068M we conclude that $g_1, \ldots, g_n$ is a regular sequence in $P \otimes_A B$.

At this point we have verified all the assumptions of Lemma 2.1 above with P, $f_1, \ldots, f_n$, and $g_i \in P \otimes_A B$ as above. In particular the annihilator J of I is freely generated by one element δ over B. Set $f_{ij} = \partial f_i/\partial x_j \in A[x_1, \ldots, x_n]$. An elementary computation shows that we can write

$$f_i(y) = f_i(z_1 + g_1, \ldots, z_n + g_n) = f_i(z) + \sum\nolimits_j f_{ij}(z) g_j + \sum\nolimits_{j,j'} F_{ijj'} g_j g_{j'}$$

for some $F_{ijj'} \in A[y_1, \ldots, y_n, z_1, \ldots, z_n]$. Taking the image in $P \otimes_A B$ the terms $f_i(z)$ map to zero and we obtain

$$f_i \otimes 1 = \sum\nolimits_j \left(1 \otimes f_{ij} + \sum\nolimits_{j'} F_{ijj'} g_{j'}\right) g_j$$

Thus we conclude from Lemma 2.1 that $\delta = \det(g_{ij})$ with $g_{ij} = 1 \otimes f_{ij} + \sum_{j'} F_{ijj'} g_{j'}$. Since $g_{j'}$ maps to zero in B, we conclude that the image of $\det(\partial f_i/\partial x_j)$ in B generates the Noether different of B over A. □

Lemma 2.3. *Let $f : Y \to X$ be a morphism of Noetherian schemes. If f satisfies the equivalent conditions of Lemma 1.2 then the different $\mathfrak{D}_f$ of f is the Kähler different of f.*

Proof. By Lemmas 0BW6 and 1.5 the different of f affine locally is the same as the Noether different. Then the lemma follows from the computation of the Noether different and the Kähler different on standard affine pieces done in Lemmas 0BVZ and 2.2. □

Lemma 2.4. *Let A be a ring. Let $n \geq 1$ and $h, f_1, \ldots, f_n \in A[x_1, \ldots, x_n]$. Set $B = A[x_1, \ldots, x_n, 1/h]/(f_1, \ldots, f_n)$. Assume that B is quasi-finite over A. Then there is an isomorphism $B \to \omega_{B/A}$ mapping $\det(\partial f_i/\partial x_j)$ to $\tau_{B/A}$.*

Proof. Let J be the annihilator of $\mathrm{Ker}(B \otimes_A B \to B)$. By Lemma 2.2 the map $A \to B$ is flat and J is a free B-module with generator ξ mapping to $\det(\partial f_i/\partial x_j)$ in B. Thus the lemma follows from Lemma 0BVT and the fact (Lemma 1.5) that $\omega_{B/A}$ is an invertible B-module. (Warning: it is necessary to prove $\omega_{B/A}$ is invertible because a finite B-module M such that $\mathrm{Hom}_B(M, B) \cong B$ need not be free.) □

Proof. Let us reformulate the statement of the proposition. Consider the category $\mathcal{C}$ whose objects, denoted Y/X, are locally quasi-finite syntomic morphism $Y \to X$ of locally Noetherian schemes and whose morphisms $b/a : Y'/X' \to Y/X$ are commutative diagrams

$$\begin{array}{ccc} Y' & \xrightarrow{b} & Y \\ \downarrow & & \downarrow \\ X' & \xrightarrow{a} & X \end{array}$$

which induce an isomorphism of Y' with an open subscheme of $X' \times_X Y$. The proposition means that for every object Y/X of $\mathcal{C}$ we have an isomorphism $c_{Y/X} : \det(NL_{Y/X}) \to \omega_{Y/X}$ with $c_{Y/X}(\delta(NL_{Y/X})) = \tau_{Y/X}$ and for every morphism $b/a : Y'/X' \to Y/X$ of $\mathcal{C}$ we have $b^* c_{Y/X} = c_{Y'/X'}$ via the identifications $b^* \det(NL_{Y/X}) = \det(NL_{Y'/X'})$ and $b^* \omega_{Y/X} = \omega_{Y'/X'}$ described above.

Given Y/X in $\mathcal{C}$ and $y \in Y$ we can find an affine opens $V \subset Y$ and $U \subset X$ with $y \in V$ and $f(V) \subset U$ such that there exists some isomorphism

$$\det(NL_{Y/X})|_V \longrightarrow \omega_{Y/X}|_V$$

mapping $\delta(NL_{Y/X})|_V$ to $\tau_{Y/X}|_V$. This follows from picking affine opens as in Lemma 1.2 part (5), the affine local description of $\delta(NL_{Y/X})$ in Remark 1.1, and Lemma 2.4. If the annihilator of the section $\tau_{Y/X}$ is zero, then these local maps are unique and automatically glue. Hence if the annihilator of $\tau_{Y/X}$ is zero, then there is a unique isomorphism $c_{Y/X} : \det(NL_{Y/X}) \to \omega_{Y/X}$ with $c_{Y/X}(\delta(NL_{Y/X})) = \tau_{Y/X}$. If $b/a : Y'/X' \to Y/X$ is a morphism of $\mathcal{C}$ and the annihilator of $\tau_{Y'/X'}$ is zero as well, then $b^* c_{Y/X}$ is the unique isomorphism $c_{Y'/X'} : \det(NL_{Y'/X'}) \to \omega_{Y'/X'}$ with $c_{Y'/X'}(\delta(NL_{Y'/X'})) = \tau_{Y'/X'}$. This follows formally from the fact that $b^* \delta(NL_{Y/X}) = \delta(NL_{Y'/X'})$ and $b^* \tau_{Y/X} = \tau_{Y'/X'}$.

We can summarize the results of the previous paragraph as follows. Let $\mathcal{C}_{nice} \subset \mathcal{C}$ denote the full subcategory of Y/X such that the annihilator of $\tau_{Y/X}$ is zero. Then we have solved the problem on $\mathcal{C}_{nice}$. For Y/X in $\mathcal{C}_{nice}$ we continue to denote $c_{Y/X}$ the solution we've just found.

Consider morphisms

$$Y_1/X_1 \xleftarrow{b_1/a_1} Y/X \xrightarrow{b_2/a_2} Y_2/X_2$$

in $\mathcal{C}$ such that Y_1/X_1 and Y_2/X_2 are objects of $\mathcal{C}_{nice}$. **Claim.** $b_1^* c_{Y_1/X_1} = b_2^* c_{Y_2/X_2}$. We will first show that the claim implies the proposition and then we will prove the claim.

Let $d, n \geq 1$ and consider the locally quasi-finite syntomic morphism $Y_{n,d} \to X_{n,d}$ constructed in Example 1.6. Then $Y_{n,d}$ is an irreducible regular scheme and the morphism $Y_{n,d} \to X_{n,d}$ is locally quasi-finite syntomic and étale over a dense open, see Lemma 1.7. Thus $\tau_{Y_{n,d}/X_{n,d}}$ is nonzero for example by Lemma 0BW9. Now

a nonzero section of an invertible module over an irreducible regular scheme has vanishing annihilator. Thus $Y_{n,d}/X_{n,d}$ is an object of $\mathcal{C}_{nice}$.

Let Y/X be an arbitrary object of $\mathcal{C}$. Let $y \in Y$. By Lemma 1.8 we can find $n, d \geq 1$ and morphisms

$$Y/X \leftarrow V/U \xrightarrow{b/a} Y_{n,d}/X_{n,d}$$

of $\mathcal{C}$ such that $V \subset Y$ and $U \subset X$ are open. Thus we can pullback the canonical morphism $c_{Y_{n,d}/X_{n,d}}$ constructed above by b to V. The claim guarantees these local isomorphisms glue! Thus we get a well defined global isomorphism $c_{Y/X} : \det(NL_{Y/X}) \to \omega_{Y/X}$ with $c_{Y/X}(\delta(NL_{Y/X})) = \tau_{Y/X}$. If $b/a : Y'/X' \to Y/X$ is a morphism of $\mathcal{C}$, then the claim also implies that the similarly constructed map $c_{Y'/X'}$ is the pullback by b of the locally constructed map $c_{Y/X}$. Thus it remains to prove the claim.

In the rest of the proof we prove the claim. We may pick a point $y \in Y$ and prove the maps agree in an open neighbourhood of y. Thus we may replace Y_1, Y_2 by open neighbourhoods of the image of y in Y_1 and Y_2. Therefore we may assume Y, X, Y_1, X_1, Y_2, X_2 are affine, say they are the spectra of rings B, A, B_1, A_1, B_2, A_2. Picture

$$\begin{array}{ccccc} B_1 & \longrightarrow & B & \longleftarrow & B_2 \\ \uparrow & & \uparrow & & \uparrow \\ A_1 & \longrightarrow & A & \longleftarrow & A_2 \end{array}$$

By assumption the spectrum of B is an affine open of both the spectrum of $A \otimes_{A_1} B_1$ and $A \otimes_{A_2} B_2$. Shrinking more we may assume there exist elements $g_i \in A \otimes_{A_i} B_i$ such that our maps give isomorphisms $(A \otimes_{A_i} B_i)_{g_i} = B$, see Properties, Lemma 0H9B. Let x_α, y_β be a sufficiently large collection of variables such that we may choose surjections

$$A_1' = A_1[x_\alpha] \to A \quad \text{and} \quad A_2' = A_2[y_\beta] \to A$$

of A_1 and A_2-algebras. Then we can choose lifts $h_i \in A_i' \otimes_{A_i} B_i$ of $g_i \in A \otimes_{A_i} B_i$ and we consider the diagram

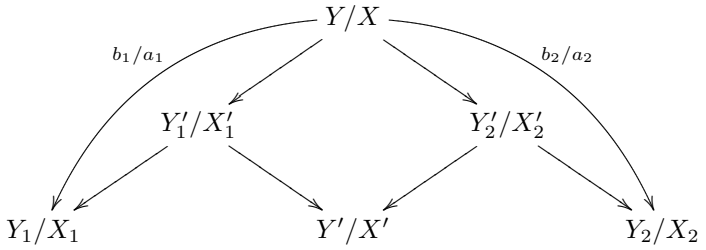
$$\begin{array}{ccccc} (A_1' \otimes_{A_1} B_1)_{h_1} & \longrightarrow & B & \longleftarrow & (A_2' \otimes_{A_1} B_2)_{h_2} \\ \uparrow & & \uparrow & & \uparrow \\ A_1' & \longrightarrow & A & \longleftarrow & A_2' \end{array}$$

By construction the two squares are cocartesian. Next, we consider the ring map

$$A' = A_1' \times_A A_2' \longrightarrow B' = (A_1' \otimes_{A_1} B_1)_{h_1} \times_B (A_2' \otimes_{A_1} B_2)_{h_2}$$

By More on Algebra, Lemma 07RU we have $A_1' \otimes_{A'} B' = (A_1' \otimes_{A_1} B_1)_{h_1}$ and $A_2' \otimes_{A'} B' = (A_2' \otimes_{A_2} B_2)_{h_2}$. In particular the fibres of the morphism $Y' = \mathrm{Spec}(B') \to \mathrm{Spec}(A') = X'$ are open subschemes of base changes of the fibres of the maps $Y_i \to X_i$, hence finite and local complete intersections (see Lemma 1.2). By More on Algebra, Lemma 08KQ the ring map $A' \to B'$ is flat and of finite presentation. Thus by Lemma 1.2 part (3) we conclude that $Y' \to X'$ is locally quasi-finite and

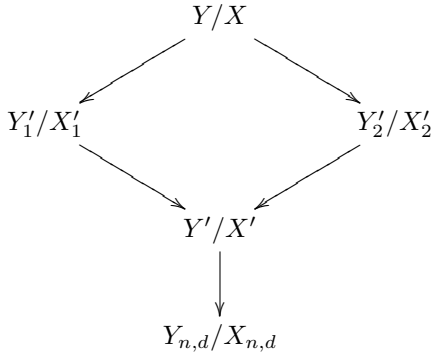
syntomic. Thus we obtain the commutative diagram



where Y_i'/X_i' corresponds to $A_i' \to (A_i' \otimes_{A_i} B_i)_{h_i}$.

In this paragraph, we fix the issue that A' and A_i' may not be Noetherian by a limit argument; we suggest the reader skip this. We can find a finitely generated $\mathbf{Z}$-subalgebra $A'' \subset A'$ and a quasi-finite and syntomic ring map $A'' \to B''$ such that $B' = A' \otimes_{A''} B''$. This follows from Limits, Lemmas 01ZM, 0C3L, and 094M. Then the ring maps $A'' \to A_1' = A_1[x_\alpha]$ will factor through a finite polynomial subalgebra $A_i[x_1, \ldots, x_n$ and the ring map $A'' \to A_2' = A_2[y_\beta]$ will factor through $A_2[y_1, \ldots, y_m]$. After increasing the sets of variables, we may also assume that $h_1 \in A_1[x_1, \ldots, x_n] \otimes_{A_1} B_1$ and $h_2 \in A_2[y_1, \ldots, y_m] \otimes_{A_2} B_2$. Increasing the sets of variables one more time, we can assume that the maps $B'' \to (A_i' \otimes_{A_i} B_i)_{h_i}$ factor through $(A_1[x_1, \ldots, x_n] \otimes_{A_1} B_1)_{h_1}$ and $(A_2[y_1, \ldots, y_m] \otimes_{A_2} B_2)_{h_2}$. Replacing Y', X', Y_1', X_1', Y_2', X_2' by the spectra of B'', A'', $(A_1[x_1, \ldots, x_n] \otimes_{A_1} B_1)_{h_1}$, $A_1[x_1, \ldots, x_n]$, $(A_2[y_1, \ldots, y_m] \otimes_{A_2} B_2)_{h_2}$, $A_2[y_1, \ldots, y_m]$, we see that we get a diagram as above where all schemes are Noetherian.

Note that Y_i'/X_i' is an object of $\mathcal{C}_{nice}$ as it obtained by taking an open subscheme of the product of an affine space with Y_i/X_i. In particular, the pullback of c_{Y_i/X_i} via (b_i/a_i) is the same as the pullback of $c_{Y_i'/X_i'}$ via $Y/X \to Y_i'/X_i'$. Now we would be done if Y'/X' is an object of $\mathcal{C}_{nice}$ (but this is likely not the case). Namely, then pulling back $c_{Y'/X'}$ around the two sides of the square, we would obtain the desired conclusion. To get around the problem that Y'/X' is not in $\mathcal{C}_{nice}$ we note the arguments above show that, after possibly shrinking all of the schemes $X, Y, X_1', Y_1', X_2', Y_2', X', Y'$ we can find some $n, d \geq 1$, and extend the diagram like so:



and then we can use the already given argument by pulling back from $c_{Y_{n,d}/X_{n,d}}$. This finishes the proof. □

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